# Marine non-predatory fish (sardine, anchovy, salmon, cod, herring, mackerel): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

## Abstract

The Heavy Metal Tested & Certified standard for marine non-predatory fish — sardine, anchovy, salmon, cod, herring, mackerel — as fillet or muscle meat, fresh or frozen, stated as consumed. These lower-trophic species carry the lower government mercury ceiling; every limit is the strictest maximum level any credible government sets for the product.

## 1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default, 8:1 basis, read-across)

## 2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

| ANALYTE | HMTC LIMIT | BASIS |
|---|---|---|
| Lead (Pb) | 200 | government maximum level (FJ) |
| Cadmium (Cd) | 50 | government maximum level (GSO) |
| Total arsenic (tAs) | 1,000 | government maximum level (Paraguay) |

| | | |
|---|---|---|
| Inorganic arsenic (iAs) | 100 | government maximum level (European Union) |
| Mercury, total (tHg) | 300 | government maximum level (EU 2023/915) |
| Methylmercury (MeHg) | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | 2,000 | read-across (EU 2024/1987) |
| Tin (Sn) | 200,000 | read-across (EU 2023/915) |
| Aluminium (Al) | 100,000 | read-across (China) |
| Total chromium (Cr) | 1,000 | government maximum level (Hong Kong) |
| Hexavalent chromium (Cr-VI) | 10 | read-across (CA SWRCB) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 200 | government maximum level (FJ) |
| Cadmium (Cd) | 50 | government maximum level (GSO) |
| Total arsenic (tAs) | 1,000 | government maximum level (Paraguay) |
| Inorganic arsenic (iAs) | 100 | government maximum level (European Union) |
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| Hexavalent chromium (Cr-VI) | 10 | read-across (CA SWRCB) |

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## 4 | Cite this standard

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## References

[1] FJ
Binds Pb.
MAXIMUM LEVEL

[2] GSO
Binds Cd.
MAXIMUM LEVEL

[3] Paraguay
Binds tAs.
MAXIMUM LEVEL

[4] EU 2023/915
Binds tHg, Sn.
MAXIMUM LEVEL

[5] EU 2024/1987
Binds Ni.
READ-ACROSS

[6] Hong Kong
Binds Cr.
MAXIMUM LEVEL

[7] CA SWRCB
Binds Cr-VI.
READ-ACROSS

[8] China
Binds Al.
READ-ACROSS

[9] European Union
Binds iAs.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

## COMPETING INTERESTS

This standard is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

## DATA AVAILABILITY

The works this standard draws on are listed in full in the references above and on the standard page at https://heavymetalcertified.com/standards/fish-marine-non-predatory. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

## PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/fish-marine-non-predatory.

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